ATTRACTORS FOR NONLINEAR REACTION-DIFFUSION EQUATIONS IN UNBOUNDED DOMAINS

by

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ABSTRACT

This paper is concerned with the existence of global attractors for a nonlinear Reaction Diffusion equation defined in unbounded domains. Under no restrictions on the space dimension or the growth order of nonlinearity, we show that the equation has an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor when the external term belongs to $L^2(\mathbf{R^n})$. Furthermore, the existence of an $(L^2(\mathbf{R^n}), L^{2p-2}(\mathbf{R^n}))$-global attractor for the equation is also established.

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1 INTRODUCTION

In this section, we explicitly state the nonlinear Reaction-Diffusion problem that we will be studying and discuss some previous results. Then we provide a brief overview of the argument used to prove the main result of this paper. Last, we discuss the organization and notation of this paper.

1.1 The Nonlinear Reaction Diffusion Problem

Consider the following Reaction Diffusion problem

$$\frac{\partial u}{\partial t} - \Delta u + \lambda u = f(x,u) + g(x), \quad x \in \mathbf{R^n}, \quad t > 0, \tag{1.1}$$

with the initial condition

$$u(x,0) = u_0(x), \quad x \in \mathbf{R^n}, \tag{1.2}$$

where λ is a positive constant, $g \in L^2(\mathbf{R^n})$ is given and f is a nonlinear function satisfying, $\forall x \in \mathbf{R^n}$ and $\forall s \in \mathbf{R}$,

$$f(x,s)s \leq -\alpha_1 |s|^p + \phi_1(x), \quad p \geq 2, \tag{1.3}$$

$$|f(x,s)| \leq \alpha_2 |s|^{p-1} + \phi_2(x), \tag{1.4}$$

$$\frac{\partial f}{\partial s}(x,s) \leq \beta, \tag{1.5}$$

$$|\frac{\partial f}{\partial s}(x,s)| \leq \alpha_3 |s|^{p-2} + \gamma, \tag{1.6}$$

where p is called the growth order of the nonlinearity, $\alpha_1,\ \alpha_2,\ \alpha_3,\ \beta$ and γ are positive constants, $\phi_1 \in L^1(\mathbf{R^n}) \cap L^\infty(\mathbf{R^n})$, and $\phi_2 \in L^2(\mathbf{R^n}) \cap L^q(\mathbf{R^n})$ with $\frac{1}{p} + \frac{1}{q} = 1$. Additionally, we let $F(x,s) = \int_0^s f(x,\tau)d\tau$ and assume that F satisfies

$$-\phi_4(x) - \alpha_4 |s|^p \leq F(x,s) \leq -\alpha_5 |s|^p + \phi_3(x), \tag{1.7}$$

where α_4 and α_5 are positive constants and $\phi_3, \phi_4 \in L^1(\mathbf{R^n})$.

This paper is concerned with the long time behavior of solutions of problem (1.1)-(1.2) defined on the unbounded domain $\mathbf{R^n}$ under the assumptions (1.3)-(1.7). Particularly, we will prove the existence of a global attractor for the dynamical system associated with problem (1.1)-(1.2) in an appropriate phase space. This global attractor is a compact set which is invariant and attracts all solutions uniformly with respect to bounded initial data. In the case of bounded domains, such attractors have been studied extensively in the literature, see, e.g., [4, 11, 13, 20, 21, 24, 25, 29] and the references therein. When the Reaction-Diffusion equations are defined in unbounded domains, the global attractors have been investigated in [1, 2, 3, 16, 17, 18, 22, 23, 26, 28]. Notice that the unboundedness of a domain introduces a major obstacle for proving the existence of an attractor because Sobolev embeddings are no longer compact in this case and so the asymptotic compactness of solutions cannot be obtained by the standard method. This difficulty can be overcome by the energy equation approach, which was introduced by Ball in [6, 7] in bounded domains and then extended to the case of unbounded domains by several authors in [8, 9, 10, 12, 14, 15, 19, 27]. The energy equation method is particularly effective when an energy functional does not involve the nonlinearity of partial differential equations under consideration. For the Reaction-Diffusion equation (1.1), the energy functional must contain the nonlinear term, and hence some new ideas must be devised for proving the asymptotic compactness of solutions. In this paper, we provide uniform estimates on the tails of solutions to circumvent the difficulty caused by the unboundedness of the domain. This idea was developed in [26] for Reaction-Diffusion equations on $\mathbf{R^n}$ and later used in several other

works, see, e.g., [1, 2, 3, 16, 17, 18, 22, 23, 28].

We remark that the existence of an $(L^2(\mathbf{R^n}), H^1(\mathbf{R^n}))$-global attractor for the Reaction-Diffusion equation on $\mathbf{R^n}$ was established in [28] recently. This attractor is compact in $H^1(\mathbf{R^n})$ and attracts all solutions with respect to the norm of $H^1(\mathbf{R^n})$ when time goes to infinity. In this paper, we will strengthen the result and prove the existence of an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor, meaning that the attractor is compact in $H^2(\mathbf{R^n})$ and attracts all solutions with respect to the norm of $H^2(\mathbf{R^n})$ when time goes to infinity. Notice that when g belongs to $L^2(\mathbf{R^n})$, the $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor is optimal in the sense that the equation has no any attractor which is more regular than $H^2(\mathbf{R^n})$ in this case.

1.2 Outline of Main Argument

As mentioned earlier, the purpose of this paper is to prove the existence of an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor for the Reaction-Diffusion equations on $\mathbf{R^n}$ under the assumptions (1.3)-(1.7). To this end, we must find a bounded absorbing set for problem (1.1)-(1.2) in $H^2(\Omega)$. This will be accomplished by deriving uniform estimates on solutions when $t \to \infty$. We also need to establish the asymptotic compactness of solutions in $H^2(\mathbf{R^n})$, which is a key step for proving the existence of an attractor in $H^2(\mathbf{R^n})$. Notice that if the external term g belongs to $L^2(\mathbf{R^n})$ only, the optimal regularity of solutions is $H^2(\mathbf{R^n})$. In this case, the asymptotic compactness of solutions can not be obtained by Sobolev embeddings even in bounded domains, because no higher regularity of solutions is available. To solve this problem, we rewrite (1.1) as

follows

$$\Delta u = \lambda u + u_t - f(x,u) - g(x). \tag{1.8}$$

We first prove the asymptotic compactness of solutions in $L^2(\mathbf{R^n})$, which is easier to get. Then the asymptotic compactness in $H^2(\mathbf{R^n})$ will follow from the asymptotic compactness of Δu in $L^2(\mathbf{R^n})$ immediately. In order to obtain such asymptotic compactness for Δu, by (1.8) we must show that every term on the right-hand side of (1.8) is asymptotically compact in $L^2(\mathbf{R^n})$. To prove the asymptotic compactness of the first two terms on the right-hand side of (1.8), we must establish tail estimates of this form; for any $\epsilon > 0$ there are two positive numbers N and T such that, for all $k \geq N$ and $t \geq T$,

$$\int_{|x|\geq k} |h(x,t)|^2 dx \leq \epsilon,$$

where $h(x,t) = u(x,t)$ or $u_t(x,t)$. Additionally, we also need to derive uniform estimates of this form; for some positive numbers T and M:

$$\int_{\mathbf{R^n}} (|h(x,t)|^2 + |\nabla h(x,t)|^2) dx \leq M, \quad \text{for all} \;\; t \geq T,$$

where $h(x,t) = u(x,t)$ or $u_t(x,t)$. These two types of estimates will be combined to prove the asymptotic compactness of the first two terms on the right-hand side of (1.8) in $L^2(\mathbf{R^n})$.

It is more difficult to prove convergence of the nonlinear term, $f(x,u)$, in $L^2(\mathbf{R^n})$. To get the desired compactness, we first derive both tail and uniform estimates for $u(x,t)$ in the space $L^{2p-2}(\mathbf{R^n})$. We then use these two estimates to prove the asymptotic compactness of solutions in $L^{2p-2}(\mathbf{R^n})$. Finally, we use the asymptotic compactness of solutions in $L^{2p-2}(\mathbf{R^n})$ and (1.6) to prove that the nonlinear term is asymptotically compact in $L^2(\mathbf{R^n})$. In the last section

of this paper, we show that these estimates also imply the existence of an $(L^2(\mathbf{R^n}), L^{2p-2}(\mathbf{R^n}))$-global attractor for the problem. Next, we introduce the organization and the notation that will be used in the remainder of this paper.

1.3 Notation and Organization of Paper

The rest of this paper is organized as follows. In the next section, we recall some fundamental concepts and results from the theory of attractors for semigroups defined in two metric spaces X and Y. In the same section, we provide a necessary and sufficient condition for the existence of an (X, Y)-global attractor. In Section 3, we establish various uniform estimates for the problem. In Section 4, we derive several uniform tail estimates of solutions when t is sufficiently large. In Section 5, we prove the asymptotic compactness of solutions in $H^2(\mathbf{R^n})$. Finally, in the last section we prove the existence of global attractors in $H^2(\mathbf{R^n})$ and $L^{2p-2}(\mathbf{R^n})$.

For the remainder of this paper, we will use the following notation. The norm of $L^2(\mathbf{R^n})$ is denoted by $\|\cdot\|$ which is defined by means of the usual inner product $(\cdot, \cdot)$. The norm for any Banach space X is written as $\|\cdot\|_X$. In particular, $\|\cdot\|_p$ denotes the norm of $L^p(\mathbf{R^n})$. The letter c is a generic positive constant which may change its value from line to line.

2 PRELIMINARIES

In this section, we introduce the definition of a semigroup that involves two spaces. We also discuss the concepts of asymptotic compactness, bounded absorbing sets and global attractors for such semigroups. Finally, we recall a necessary and sufficient condition for the existence of a global attractor for a semigroup involving two spaces. The following definitions are similar to standard ones in [11, 20, 21, 24] and can be also found in [25].

2.1 Semigroups

Hereafter, we assume that X is a complete metric space with distance $d_X(\cdot,\cdot)$.

Definition 2.1. A **semigroup on** X is a family of mappings $\{S(t)\}_{t\geq 0}$ such that:

(i) $S(t) : X \to X \qquad \forall t \geq 0$

(ii) $S(t+\tau) = S(t)S(\tau) \qquad \forall t, \tau \geq 0$

(iii) $S(0) = I_X$ (Identity operator on X).

The space X is generally known as the phase space of the semigroup. Next let (X, d_X) and (Y, d_Y) be complete metric spaces. Then we can define a generalization of a semigroup on X.

Definition 2.2. A semigroup, $\{S(t)\}_{t\geq 0}$, on X is said to be an (X, Y)**-semigroup** if the following conditions hold:

(i) $X \cap Y \neq \emptyset$.

(ii) $S(t)(X) \subset X \cap Y \ \forall t > 0$.

Note that if $X = Y$, then an (X, Y)-semigroup reduces to a standard semigroup on X. In this sense, the concept of (X, Y)-semigroups is an extension of semigroups on a single space. In practice, the semigroup $\{S(t)\}_{t\geq 0}$ is actually a family of solution operators for a given ordinary or partial differential equation. In this case, the study of long time behavior of solutions for the equation amounts to studying the limiting behavior of the corresponding semigroup as time goes to infinity. In many cases, the long time behavior of a semigroup is governed by a compact invariant set which attracts all solutions. Such a set is called a global attractor for the semigroup. The definition of a global attractor and related concepts are presented below.

2.2 Global Attractors

Definition 2.3. Let $\{S(t)\}_{t\geq 0}$ be an (X, Y)-semigroup and $\mathbb{A}$ be a subset of $X \cap Y$. The set $\mathbb{A}$ is called an (X, Y)**-global attractor of** $\{S(t)\}_{t\geq 0}$ if the following conditions are satisfied:

(i) $\mathbb{A}$ is compact in (Y, d_Y),

(ii) $\mathbb{A}$ is invariant, i.e., $S(t)\mathbb{A} = \mathbb{A}$ for every $t \geq 0$,

(iii) $\mathbb{A}$ attracts every bounded subset of E of (X, d_X) with respect to the metric on (Y, d_Y), i.e., $\quad d_Y(S(t)E, \mathbb{A}) \to 0$ as $t \to \infty$,

where $d_Y(S(t)E, \mathbb{A}) = \sup\limits_{e\in E} \inf\limits_{a\in\mathbb{A}} d_Y(S(t)e, a)$.

Notice that the global attractor for a semigroup is unique if it exists. By definition, it is easy to verify that the global attractor contains all steady solutions as well as periodic solutions of differential equations. In fact, the

global attractor contains all dynamical information of a semigroup. This is why the concept of the global attractor is so important in the field of dynamical systems, and the investigation of existence of such an attractor is crucial for understanding the dynamics of dissipative equations.

2.3 Asymptotic Compactness and Bounded Absorbing Sets

We now present a necessary and sufficient condition for the existence of a global attractor for a semigroup. To this end, we first need to introduce the concepts of bounded absorbing sets and asymptotic compactness.

Definition 2.4. Let $\{S(t)\}_{t\geq 0}$ be an (X, Y)-semigroup and B be a subset of Y. The set B is said to be an (X, Y)**-absorbing set for** $\{S(t)\}_{t\geq 0}$ if for every bounded subset E of (X, d_X) there exist a positive number T depending on E such that

$$S(t)E \subset B, \quad \forall t \geq T.$$

Additionally, if B is bounded in (Y, d_Y) then B is called a **bounded** (X, Y)**-absorbing set for** $\{S(t)\}_{t\geq 0}$.

Definition 2.5. Let $\{S(t)\}_{t\geq 0}$ be an (X, Y)-semigroup. $\{S(t)\}_{t\geq 0}$ is said to be (X, Y)**-asymptotically compact** if $t_n \to \infty$ and $\{x_n\}$ is bounded in both (X, d_X) and (Y, d_Y) imply that $\{S(t_n)x_n\}_{n=1}^{\infty}$ has a convergent subsequence in (Y, d_Y).

Definition 2.6. Let (X, d_X) and (Y, d_Y) be metric spaces. (X, d_X) and (Y, d_Y) are said to be **consistent** if for any $\{z_n\}_{n=1}^{\infty} \subset X \cap Y$, $x \in X$ and $y \in Y$, $z_n \to x$ in (X, d_X) and $z_n \to y$ in (Y, d_Y) imply that $x = y$.

Although the following theorem has been proved in [25], we reproduce a proof of the result for clarity and completeness.

Theorem 2.7. *Suppose (X, d_X) and (Y, d_Y) are consistent and $\{S(t)\}_{t\geq 0}$ is an (X, Y)-semigroup. Let $\mathbb{A}$ be an (X, X)-global attractor of $\{S(t)\}_{t\geq 0}$. $\mathbb{A}$ is also an (X, Y)-global attractor if and only if the following conditions hold:*

1. *$\{S(t)\}_{t\geq 0}$ has a bounded (X, Y)-absorbing set,*
2. *$\{S(t)\}_{t\geq 0}$ is (X, Y)-asymptotically compact.*

Proof. ($\Leftarrow$) Assume that $\{S(t)\}_{t\geq 0}$ is (X, Y)-asymptotically compact and has a bounded (X, Y)-absorbing set, B. $\mathbb{A}$ is bounded in X since it is, by assumption, compact in X. By the definition of (X, Y)-absorbing set we find that there exists a $T \geq 0$ such that

$$S(t)\mathbb{A} \subseteq B, \qquad \forall t \geq T.$$

By the invariance of $\mathbb{A}$ we find that $\mathbb{A} \subset B$. Next we let $E = \{x \in X : d_X(x, \mathbb{A}) < 1\}$ and $B_1 = B \cap E$. Then we let $\omega(B_1)$ be the ω-limit set of B_1 in (Y, d_Y),

$$\omega(B_1) = \{y \in Y : \exists y_n \in B_1, t_n \to \infty \text{ such that } d_Y(S(t_n)y_n, y) \to 0\}.$$

By the (X, Y)-asymptotic compactness of $\{S(t)\}_{t\geq 0}$, we can prove that $\omega(B_1)$ is compact in Y and attracts every bounded subset of X with respect to (Y, d_Y) (see, [24]). Which implies that

$$d_Y(S(t)\mathbb{A}, \omega(B_1)) \to 0, \text{ as } t \to \infty,$$

since $\mathbb{A}$ is bounded in (X, d_X). By the invariance of $\mathbb{A}$ we see that $d_Y(\mathbb{A}, \omega(B_1)) = 0$. Therefore, we find that

$$\mathbb{A} \subseteq \omega(B_1). \tag{2.9}$$

Next we let $y \in \omega(B_1)$. By definition of $\omega(B_1)$ we find that there exists a sequence y_n in B_1 and $t_n \to \infty$ such that

$$d_Y(S(t_n)y_n, y) \to 0, \text{ as } n \to \infty.$$

Moreover, since B_1 is bounded in (X, d_X) we find that, up to a subsequence, there is an $x \in \mathbb{A}$ such that

$$d_X(S(t_n)y_n, x) \to 0, \text{ as } n \to \infty.$$

Then since X and Y are consistent metric spaces we find that $y = x \in \mathbb{A}$. Therefore, since y was arbitrary we find that

$$\omega(B_1) \subseteq \mathbb{A}. \tag{2.10}$$

Combining (2.9) and (2.10) we find that $\omega(B_1) = \mathbb{A}$; therefore, the proof of sufficiency is complete.

($\Rightarrow$) First assume that $\mathbb{A}$ is a (X, Y)-global attractor. Then let F be defined as

$$F = \{y \in Y : d_Y(y, \mathbb{A}) < 1\}.$$

F is a bounded (X, Y)-absorbing set since $\mathbb{A}$ attracts all bounded sets in (X, d_X) with respect to the metric (Y, d_Y).

Next we prove the asymptotic compactness of the semigroup. Let $t_n \to \infty$ and x_n be a bounded sequence in (X, d_X). The attracting property and the

compactness of $\mathbb{A}$ imply that there is a sequence z_n in $\mathbb{A}$ such that,

$$d_Y(S(t_n)x_n, z_n) = d_Y(S(t_n)x_n, \mathbb{A}).$$

The right hand side of the previous inequality converges to 0 because $\mathbb{A}$ is a (X, Y)-global attractor; therefore, we find that

$$d_Y(S(t_n)x_n, z_n) \to 0 \text{ as } n \to \infty. \tag{2.11}$$

Moreover, there is a $z \in \mathbb{A}$ such that, up to a subsequence, $z_n \to z$ in (Y, d_Y) since $\mathbb{A}$ is compact in (Y, d_Y). By (2.11) we find that, up to a subsequence, $S(t_n)x_n \to z$ in (Y, d_Y). Therefore, $\{S(t)\}_{t\geq 0}$ is (X, Y)-asymptotically compact and has a bounded (X, Y)-absorbing set. The proof is completed. □

Remark 2.1. In [6, 7] the author proves that a semigroup has an (X, X)-global attractor if and only if the semigroup is (X, X)-asymptotically compact and the semigroup has a bounded (X, X)-absorbing set. We will use this result to verify the existence of an $(L^2(\mathbf{R^n}), L^2(\mathbf{R^n}))$-global attractor. This result is needed because Theorem 2.9 requires the existence of an $(L^2(\mathbf{R^n}), L^2(\mathbf{R^n}))$-global attractor in order to prove the existence of an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor.

In the next sections, we establish estimates that are necessary to apply Theorem 2.7 and prove that the solution of (1.1)-(1.7) has an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor.

3 UNIFORM ESTIMATES

In this section, we derive uniform estimates of solutions for problem (1.1)-(1.2) under the assumptions (1.3)-(1.7). These estimates are needed for proving the existence of a bounded absorbing set and asymptotic compactness of the semigroup associated with problem (1.1)-(1.2). We start with the estimates in $L^2(\mathbf{R^n})$.

Lemma 3.1. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the solution u satisfies*

$$\|u(t)\| \leq M, \quad \forall t \geq T,$$

where M depends on $(\lambda, f, g); T$ depends on (λ, f, g) and R when $\|u_0\| \leq R$.

Proof. First we take the inner product of (1.1) with u in $L^2(\mathbf{R^n})$ to obtain the equation

$$\frac{1}{2}\frac{d}{dt}\|u\|^2 + \|\nabla u\|^2 + \lambda\|u\|^2 = \int_{\mathbf{R^n}} fudx + (g, u). \tag{3.12}$$

Next we bound the right hand side of (3.12). By (1.3) we get

$$\int_{\mathbf{R^n}} fudx \leq -\alpha_1 \int_{\mathbf{R^n}} |u|^p + \int_{\mathbf{R^n}} \phi_1 dx, \quad p \geq 2.$$

Then since $\phi_1 \in L^1(\mathbf{R^n})$ we find that

$$\int_{\mathbf{R^n}} fudx \leq -\alpha_1\|u\|_p^p + c. \tag{3.13}$$

Additionally, since $g \in L^2$ we find that

$$|(g, u)| \leq \|g\|\,\|u\| = c\|u\|.$$

Applying Young's inequality to the right-hand side of the above we obtain

$$(g,u) \leq \frac{\lambda}{2}\|u\|^2 + \frac{1}{2\lambda}c^2. \tag{3.14}$$

Using (3.14) and (3.13) to bound the right hand side of (3.12) we find that

$$\frac{d}{dt}\|u\|^2 + 2\|\nabla u\|^2 + \lambda\|u\|^2 + 2\alpha_1\|u\|_p^p \leq c. \tag{3.15}$$

Since $\|\nabla u\|^2$ is non-negative we find that

$$\frac{d}{dt}\|u\|^2 + \lambda\|u\|^2 \leq c. \tag{3.16}$$

Then we find (3.16) and Gronwall's lemma imply that

$$\|u(t)\|^2 \leq e^{-\lambda t}\|u_0\|^2 + \frac{c}{\lambda} \leq e^{-\lambda t}R^2 + \frac{c}{\lambda}, \quad \forall t \geq 0.$$

Consequently, if $T = \frac{1}{\lambda}ln|\frac{R^2\lambda}{c}|$ then $\forall t \geq T$

$$\|u(t)\|^2 \leq \frac{2c}{\lambda}, \tag{3.17}$$

hence the proof is completed. □

Next, we derive uniform estimates for u in $H^1(\mathbf{R^n})$ and $L^p(\mathbf{R^n})$.

Lemma 3.2. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the solution u satisfies*

$$\|\nabla u(t)\| + \int_{\mathbf{R^n}} |u(x,t)|^p dx \leq M, \quad \forall t \geq T,$$

and

$$\int_t^{t+1} \|u_t(x,t)\|^2 dt \leq M, \quad \forall t \geq T,$$

where M depends on (λ, f, g); T depends on (λ, f, g) and R when $\|u_0\| \leq R$.

Proof. First integrating (3.15) from t to $t+1$ we obtain

$$2\int_t^{t+1} \|\nabla u\|^2 dt + 2\alpha_1 \int_t^{t+1} \int_{\mathbf{R^n}} |u|^p dx dt \leq c + \|u(t)\|^2$$

since $\|u(t+1)\|^2$ is non-negative. Then by Lemma 3.1 we find that

$$\int_t^{t+1} \|\nabla u\|^2 dt + \alpha_1 \int_t^{t+1} \int_{\mathbf{R^n}} |u|^p dx dt \leq c, \quad \forall t \geq T. \tag{3.18}$$

Next we take the inner product of (1.1) with u_t in $L^2(\mathbf{R^n})$ to obtain the equation

$$\|u_t\|^2 + \frac{d}{dt}(\frac{1}{2}\|\nabla u\|^2 + \frac{\lambda}{2}\|u\|^2 - (g,u) - \int_{\mathbf{R^n}} F(x,u)dx) = 0. \tag{3.19}$$

Since $\|u_t\|^2$ is non-negative we find that

$$\frac{d}{dt}(\frac{1}{2}\|\nabla u\|^2 + \frac{\lambda}{2}\|u\|^2 - (g,u) - \int_{\mathbf{R^n}} F(x,u)dx) \leq 0. \tag{3.20}$$

Then we let $s \in (t, t+1)$ and integrate the previous inequality from s to $t+1$ to get the inequality

$$\frac{1}{2}\|\nabla u(t+1)\|^2 + \frac{\lambda}{2}\|u(t+1)\|^2 - (g, u(t+1)) - \int_{\mathbf{R^n}} F(x, u(t+1))dx$$
$$\leq \frac{1}{2}\|\nabla u(s)\|^2 + \frac{\lambda}{2}\|u(s)\|^2 - (g, u(s)) - \int_{\mathbf{R^n}} F(x, u(s))dx.$$

Next we integrate the previous inequality with respect to s on $(t, t+1)$ to obtain the inequality

$$\frac{1}{2}\|\nabla u(t+1)\|^2 + \frac{\lambda}{2}\|u(t+1)\|^2 - (g, u(t+1)) - \int_{\mathbf{R^n}} F(x, u(t+1))dx \leq$$
$$\int_t^{t+1} \frac{1}{2}\|\nabla u\|^2 ds + \int_t^{t+1} \frac{\lambda}{2}\|u\|^2 ds - \int_t^{t+1} (g,u)ds - \int_t^{t+1} \int_{\mathbf{R^n}} F(x,u)dxds \tag{3.21}$$

Next we bound the right-hand side of (3.21). By (1.7) and (3.18) we have

$$-\int_{\mathbf{R^n}} F(x,u)dx \le \alpha_4 \int_{\mathbf{R^n}} |u|^p dx + \int_{\mathbf{R^n}} \phi_4(x)dx \le \alpha_4 \int_{\mathbf{R^n}} |u|^p dx + c. \quad (3.22)$$

$$-\int_t^{t+1} \int_{\mathbf{R^n}} F(x,u)dxdu \le \alpha_4 \int_t^{t+1} \int_{\mathbf{R^n}} |u|^p dxdu + \int_t^{t+1} \int_{\mathbf{R^n}} \phi_4(x)dxds \le c. \quad (3.23)$$

Additionally, $|(g,u)| \le \|g\|\|u\| = \frac{1}{2}\|u\|^2 + c$; therefore by Lemma 3.1 we obtain

$$\int_t^{t+1} |(g,u)|dt \le c. \quad (3.24)$$

Next we apply (3.24) and (3.23) to (3.21) to obtain

$$\frac{1}{2}\|\nabla u(t+1)\|^2 + \frac{\lambda}{2}\|u(t+1)\|^2 - (g, u(t+1)) - \int_{\mathbf{R^n}} F(x, u(t+1))dx \le c. \quad (3.25)$$

Then by (1.7) and (3.25) we find that, for all $t \ge T$,

$$\frac{1}{2}\|\nabla u(t+1)\|^2 + \alpha_5 \int_{\mathbf{R^n}} |u(x,t+1)|^p dx \le c + (g, u(t+1)) \le c. \quad (3.26)$$

Finally, integrating (3.19) over $(t, t+1)$ and using (3.26) and (1.7) we find that, $\forall t \ge T+1$,

$$\int_t^{t+1} \|u_t(x,t)\|^2 dt \le c. \quad (3.27)$$

The lemma follows from (3.26) and (3.27); therefore, the proof is complete. □

In the next lemma, we differentiate (1.1) and take the inner product of the resulting equation with u_t in $L^2(\mathbf{R^n})$. Then we estimate $u_t(x,t)$ in $L^2(\mathbf{R^n})$ and derive an estimate needed to prove the uniform estimate for $u_t(x,t)$ in $H^1(\mathbf{R^n})$.

Lemma 3.3. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the solution u satisfies*

$$\|u_t(x,t)\| \le M, \quad \forall t \ge T,$$

and

$$\int_t^{t+1} \|\nabla u_t\|^2 dt + \int_t^{t+1} \left(\int_{\mathbf{R^n}} |u_t|^{\frac{2n}{n-2}} dx \right)^{\frac{n-2}{n}} dt \leq M, \quad \forall t \geq T,$$

where M depends on (λ, f, g); T depends on (λ, f, g) and R when $\|u_0\| \leq R$.

Proof. First we let $u_t = v$ and differentiate (1.1) with respect to t to show that v must satisfy

$$v_t - \Delta v + \lambda v = \frac{\partial f}{\partial u}(x, u)v. \tag{3.28}$$

Next we take the inner product of the previous equation with v in $L^2(\mathbf{R^n})$ and use (1.5) to obtain the inequalities:

$$\frac{1}{2}\frac{d}{dt}\|v\|^2 + \|\nabla v\|^2 + \lambda\|v\|^2 \leq \beta\|v\|^2, \tag{3.29}$$

$$\frac{1}{2}\frac{d}{dt}\|v\|^2 + \|\nabla v\|^2 \leq \beta\|v\|^2, \tag{3.30}$$

$$\frac{1}{2}\frac{d}{dt}\|v\|^2 \leq \beta\|v\|^2. \tag{3.31}$$

We then apply the Uniform Gronwall Lemma and Lemma 3.2 to (3.31) to find that,

$$\|v(t)\| = \|u_t\| \leq M, \quad \forall t > T + 1. \tag{3.32}$$

Integrating (3.30) from t to $t + 1$ we obtain the inequality

$$\|v(t+1)\|^2 + 2\int_t^{t+1} \|\nabla v\|^2 dt \leq c\|v\|^2.$$

Applying (3.32) to the previous inequality we find that, $\forall t > T + 1$

$$\int_t^{t+1} \|\nabla v\|^2 dt \leq c. \tag{3.33}$$

Finally, we find that (3.33) along with the embedding $H^1 \hookrightarrow L^{\frac{2n}{n-2}}$ imply that, $\forall t > T + 1$

$$\int_t^{t+1} \left(\int_{\mathbf{R^n}} |u_t|^{\frac{2n}{n-2}} dx \right)^{\frac{n-2}{n}} dt \leq c. \tag{3.34}$$

The main results of the lemma follow from (3.34) and (3.32); therefore, the proof is complete.

□

Next we use induction to estimate $u_t(x,t)$ in $L^r(\mathbf{R^n})$. The idea for the proof of the next lemma was initially given in [29] and used again in [25] to prove a similar result about bounded domains.

Lemma 3.4. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then for any $r \geq 2$ the solution u satisfies*

$$\int_{\mathbf{R^n}} |u_t(x,t)|^r dx \leq M, \quad \forall t \geq T,$$

where M depends on (λ, f, g) and r; T depends on $(\lambda, f, g), r$ and R when $\|u_0\| \leq R$.

Proof. We show by induction that, for every $k \geq 0$, there are positive constants M_k and T_k such that for all $t \geq T_k$,

$$\int_{\mathbf{R^n}} |u_t|^{2(\frac{n}{n-2})^k} dx \leq M_k, \tag{3.35}$$

and

$$\int_t^{t+1} \left(\int_{\mathbf{R^n}} |u_t|^{2(\frac{n}{n-2})^{k+1}} dx \right)^{\frac{n-2}{n}} dt \leq M_k. \tag{3.36}$$

First we note that Lemma 3.3 provides the base case, k=0, for the induction on (3.35) and (3.36). Next suppose that (3.35) and (3.36) are true for $k \geq 0$. Multiply (3.28) by $|v|^{2(\frac{n}{n-2})^{k+1}-2} v$ and apply (1.5) to find that

$$c_1 \frac{d}{dt} \int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx + c_2 \int_{\mathbf{R^n}} |\nabla(v^{(\frac{n}{n-2})^{k+1}})|^2 dx + \lambda \int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx$$

$$\leq \beta \int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx, \tag{3.37}$$

where c_1 and c_2 depends on n and k. Dropping two of the non-negative terms on the left side of (3.37) we obtain the inequality

$$\frac{d}{dt} \int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx \leq c \int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx. \tag{3.38}$$

Next we multiply (3.38) by $\frac{n-2}{n} \left(\int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx \right)^{-\frac{2}{n}}$ to obtain the inequality

$$\frac{d}{dt} \left(\int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx \right)^{\frac{n-2}{n}} \leq c \left(\int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx \right)^{\frac{n-2}{n}}. \tag{3.39}$$

Applying the uniform Gronwall lemma to (3.39) and use (3.36) we find that, for all $t \geq T_k + 1$,

$$\int_{\mathbf{R^n}} |v|^{2(\frac{n}{n-2})^{k+1}} dx \leq c. \tag{3.40}$$

Now we integrate (3.37) from t to t+1 and apply (3.40) to obtain that, for all $t \geq T_k + 1$,

$$\int_t^{t+1} \int_{\mathbf{R^n}} |\nabla (v^{(\frac{n}{n-2})^{k+1}})|^2 dx dt \leq c. \tag{3.41}$$

Then we find that (3.41) and the embedding $H^1 \hookrightarrow L^{\frac{2n}{n-2}}$ imply that, $\forall t > T_k + 1$,

$$\int_t^{t+1} \left(\int_{\mathbf{R^n}} |v|^{(2\frac{n}{n-2})^{k+2}} \right)^{\frac{n-2}{n}} dx dt \leq c. \tag{3.42}$$

Now we find that (3.40) and (3.42) prove the k+1 case and complete the induction proof. Next we extend the result to any $r \geq 2$. Note that the case $r = 2$ is covered by Lemma 3.3. For $r > 2$, we take an integer m large enough such that

$$2 < r < 2 \left(\frac{n}{n-2} \right)^m.$$

Then by the Gagliardo-Nirenberg inequality and (3.35) we find that

$$\|u_t(t)\|_r \leq c,$$

which completes the proof. $\square$

Next we drive the necessary uniform estimate for $u(x,t)$ in $L^{2p-2}(\mathbf{R^n})$. This result is needed to prove the asymptotic compactness of the nonlinear term in $L^2(\mathbf{R^n})$, which in turn is needed to prove the asymptotic compactness of $u(x,t)$ in $H^2(\mathbf{R^n})$.

Lemma 3.5. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the solution u satisfies*

$$\|u(t)\|_{2p-2} \leq M, \quad \forall t \geq T,$$

where M depends on the data (λ, f, g); T depends on (λ, f, g) and R when $\|u_0\| \leq R$.

Proof. First we take the inner product of (1.1) with $|u|^{p-2}u$ in $L^2(\mathbf{R^n})$ and integrate by parts to obtain the equation

$$(p-1)\int_{\mathbf{R^n}} |u|^{p-2}|\nabla u|^2 dx - \int_{\mathbf{R^n}} f(x,u)|u|^{p-2}u dx =$$

$$-\int_{\mathbf{R^n}} |u|^{p-2}uu_t dx - \lambda \int_{\mathbf{R^n}} |u|^p dx + \int_{\mathbf{R^n}} |u|^{p-2}ug dx. \tag{3.43}$$

Next we derive several inequalities to bound the nonlinear term on the left side of the previous equation. First we multiply (1.3) by $|u|^{p-2}$ and then integrate to obtain the inequality

$$\int_{\mathbf{R^n}} f(x,u)u|u|^{p-2}dx \leq -\alpha_1 \int_{\mathbf{R^n}} |u|^{2p-2}dx + \int_{\mathbf{R^n}} \phi_1(x)|u|^{p-2}dx. \tag{3.44}$$

Next we apply Hölder's inequality to the second term on the right hand side of (3.44) to find that

$$\int_{\mathbf{R^n}} \phi_1(x)|u|^{p-2}dx \leq \|\phi_1\|_r \left\||u|^{p-2}\right\|_s = \|\phi_1\|_r \|u\|_{2p-2}^{p-2}, \tag{3.45}$$

where $(r,s) = (\frac{2p-2}{p}, \frac{2p-2}{p-2})$ are conjugate pairs. Additionally, since $\phi_1 \in L^1(\mathbf{R^n}) \cap L^\infty(\mathbf{R^n})$, we can apply Hölder's inequality again to find that

$$\|\phi_1\|_r = \left(\int_{\mathbf{R^n}} |\phi_1||\phi_1|^{r-1}dx\right)^{\frac{1}{r}} \leq \|\phi_1\|_\infty^{\frac{r-1}{r}} \|\phi_1\|_1^{\frac{1}{r}} \leq c. \tag{3.46}$$

Now we find that (3.45), (3.46) and Young's inequality together imply

$$\int_{\mathbf{R^n}} \phi_1(x)|u|^{p-2}dx \leq c + \frac{1}{4}\alpha_1 \int_{\mathbf{R^n}} |u|^{2p-2}dx. \tag{3.47}$$

We are now in a position to prove the inequality needed to bound the nonlinear term on the left side of (3.43). Note that (3.47) and (3.44) imply the inequality

$$\frac{3}{4}\alpha_1 \int_{\mathbf{R^n}} |u|^{2p-2}dx \leq -\int_{\mathbf{R^n}} f(x,u)|u|^{p-2}udx + c. \tag{3.48}$$

Next we apply Young's inequality to the right hand side of (3.43) to obtain

$$-\int_{\mathbf{R^n}} |u|^{p-2}uu_t dx + \int_{\mathbf{R^n}} |u|^{p-2}ugdx \leq \frac{1}{4}\alpha_1 \int_{\mathbf{R^n}} |u|^{2p-2}dx + c(\|u_t\|^2 + \|g\|^2). \tag{3.49}$$

Then we find that (3.48), (3.49), Lemma 3.1 and Lemma 3.3 together imply that

$$(p-1)\int_{\mathbf{R^n}} |u|^{p-2}|\nabla u|^2 dx + \frac{1}{2}\alpha_1 \int_{\mathbf{R^n}} |u|^{2p-2}dx \leq c. \tag{3.50}$$

Finally, we find that lemma 3.5 follows, a fortiori, from inequality (3.50). □

Lemma 3.6. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the solution u satisfies*

$$\|\Delta u\| \leq M, \quad \forall t \geq T,$$

where M depends on the data (λ, f, g); T depends on (λ, f, g) and R when $\|u_0\| \leq R$.

Proof. First we take the inner product of (1.1) in $L^2(\mathbf{R^n})$ with Δu and rearrange terms to obtain the equation

$$\|\Delta u\|^2 + \lambda \int_{\mathbf{R^n}} |\nabla u|^2 dx = \int_{\mathbf{R^n}} u_t \Delta u dx - \int_{\mathbf{R^n}} f(x,u)\Delta u dx - \int_{\mathbf{R^n}} g \Delta u dx. \quad (3.51)$$

Now we bound the terms on the left side of of the previous equation. Note that by (1.4) and Lemma 3.5 we find that, for $t \geq T$,

$$\|f\|^2 = \int_{\mathbf{R^n}} |f(x,u(x,t))|^2 dx \leq c \left(\int_{\mathbf{R^n}} |u(x,t)|^{2p-2} dx + \int_{\mathbf{R^n}} \phi_2^2(x) dx \right) \leq c, \quad (3.52)$$

since $\phi_2 \in L^1$. Then applying Young's inequality to (3.51) we find that

$$\|\Delta u\|^2 \leq \|u_t\|^2 + \frac{1}{4}\|\Delta u\|^2 + \|f\|^2 + \frac{1}{4}\|\Delta u\|^2 + \|g\|_2 + \frac{1}{4}\|\Delta u\|^2,$$

which implies that,

$$\frac{1}{4}\|\Delta u\|^2 \leq \|u_t\|^2 + \|f\|^2 + \|g\|^2. \quad (3.53)$$

Finally, since $g \in L^2(\mathbf{R^n})$, we find that (3.52), (3.53) and Lemma 3.3 imply that, $\forall t \geq T$,

$$\|\Delta u\|^2 \leq c, \quad (3.54)$$

hence the proof is complete. □

Lastly, we derive the uniform estimates for u_t in $H^1(\mathbf{R^n})$. Lemma 3.2 and the next lemma together are needed to prove the convergence of the first two terms on the right hand side of (1.8) in $L^2(\mathbf{R^n})$.

Lemma 3.7. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the solution u satisfies*

$$\|\nabla u_t(t)\| \leq M, \quad \forall t \geq T,$$

where M depends on the data (λ, f, g); T depends on (λ, f, g) and R when $\|u_0\| \leq R$.

Proof. First we let $v = u_t$ and then take the inner product of (3.28) with $-\Delta v$ in $L^2(\mathbf{R^n})$ to obtain

$$\frac{1}{2}\frac{d}{dt}\|\nabla v\| + \|\Delta v\|^2 + \lambda\|\nabla v\| = -\int_{\mathbf{R^n}} \frac{\partial f}{\partial u}(x,u) v \Delta v dx. \tag{3.55}$$

Now we apply (1.6) to the previous equation to find that,

$$\frac{1}{2}\frac{d}{dt}\|\nabla v\| + \|\Delta v\|^2 \leq \alpha_3 \int_{\mathbf{R^n}} |u|^{p-2}|v||\Delta v| dx + \gamma \int_{\mathbf{R^n}} |v||\Delta v| dx. \tag{3.56}$$

Then we apply Hölder's inequality to (3.56) to get that

$$\frac{1}{2}\frac{d}{dt}\|\nabla v\| + \|\Delta v\|^2 \leq \|u\|_{2p-2}^{p-2}\|v\|_{2p-2}\|\Delta v\| + \gamma\|v\|\|\Delta v\|. \tag{3.57}$$

Now we apply Young's inequality, Lemma 3.4, Lemma 3.5, and Lemma 3.6 to the right side of the previous inequality to find that, $\forall t \geq T$,

$$\frac{1}{2}\frac{d}{dt}\|\nabla v\| + \|\Delta v\|^2 \leq c + \frac{1}{2}\|\Delta v\|^2. \tag{3.58}$$

Finally, we find that the uniform Gronwall lemma, Lemma 3.3 and (3.58) together imply that, $\forall t \geq T + 1$,

$$\|\nabla u_t(t)\|^2 \leq c. \tag{3.59}$$

The main result of this lemma follows from (3.59); therefore, the proof is complete. $\square$

We have now finished deriving all of the necessary uniform estimates needed to prove the asymptotic compactness of the solution operator in $H^2(\mathbf{R^n})$. In the next section, we derive all of the necessary tail estimates needed to complete the proof of the asymptotic compactness of the semigroup associated with the solution of the Reaction Diffusion equations.

4 TAIL ESTIMATES

In this section, we derive tail estimates by using a smooth cut-off function, $\rho(x)$, defined below. In the next lemma, we multiply (1.1) by $\rho^2(x)$ and then take the inner product of the resulting equation with u in $L^2(\mathbf{R^n})$. Then we use the resulting equation to derive a tail estimate for $u(x,t)$ in $L^2(\mathbf{R^n})$. The idea of the next proof can be found in [26].

Lemma 4.1. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then given $\epsilon > 0$, there are $k_0 = k_0(\epsilon)$ and $T_0 = T_0(\epsilon)$ such that the solution u satisfies*

$$\int_{|x|\geq k_0} |u(x,t)|^2 dx \leq \epsilon, \quad \forall t \geq T_0,$$

where k_0 depends on the data (λ, f, g) and ϵ; T_0 depends on (λ, f, g), ϵ and R when $\|u_0\| \leq R$.

Proof. First we choose a smooth function, $\theta : \mathbf{R}^+ \to \mathbf{R}$, such that:

$$\forall s \in \mathbf{R}^+ \quad 0 \leq \theta(s) \leq 1 \tag{4.60}$$

$$\theta(s) = 0 \quad \text{for } 0 \leq s \leq 1 \tag{4.61}$$

$$\theta(s) = 1 \quad \text{for } s \geq 2. \tag{4.62}$$

Note that (4.60)- (4.62) imply that there is a $c > 0$ such that $|\theta'(s)| + |\theta''(s)| \leq c$ for all $s \in \mathbf{R}^+$. Then we multiply (1.1) by $\rho^2(x)$, where $\rho(x) = \theta\left(\frac{|x|^2}{k^2}\right)$, and take the inner product of the resulting equation with u in $L^2(\mathbf{R^n})$ to find that,

$$\frac{d}{dt}\|u\rho\|^2 - \int_{\mathbf{R^n}} \Delta u u \rho^2 dx + \lambda \|u\rho\|^2 = \int_{\mathbf{R^n}} f(x,u) u \rho^2 + \int_{\mathbf{R^n}} g u \rho^2 dx. \tag{4.63}$$

Next we proceed to bound the terms in the previous equation. First we find that

$$-\int_{\mathbf{R^n}} \Delta u u \rho^2 dx = \int_{\mathbf{R^n}} \rho^2 |\nabla u|^2 dx + \int_{\mathbf{R^n}} \frac{4x}{k^2} \theta' \left(\frac{|x|^2}{k^2} \right) \rho u \nabla u dx. \tag{4.64}$$

Additionally, we find that Young's inequality, (4.60) and (4.62) all together imply the inequality

$$\begin{aligned} \left| \int_{\mathbf{R^n}} \frac{4x}{k^2} \theta' \left(\frac{|x|^2}{k^2} \right) \rho u \nabla u dx \right| &\leq c \int_{k \leq |x| \leq \sqrt{2}k} \frac{|x|}{k^2} |u| |\nabla u| dx \\ &\leq \frac{c}{k} \int_{k \leq |x| \leq \sqrt{2}k} |u| |\nabla u| dx \leq \frac{c}{k} (\|u\| \|\nabla u\|). \end{aligned} \tag{4.65}$$

Therefore, by Lemma 3.1, Lemma 3.6 and (4.65), given $\epsilon > 0$, there is a $k_1 = k_1(\epsilon)$ such that

$$\left| \int_{\mathbf{R^n}} \frac{4x}{k^2} \theta' \left(\frac{|x|^2}{k^2} \right) \rho u \nabla u dx \right| \leq \epsilon, \quad \forall k \geq k_1 \text{ and } \forall t \geq T. \tag{4.66}$$

Moreover, since $g \in L^2(\mathbf{R^n})$ we find that

$$\left| \int_{\mathbf{R^n}} g u \rho^2 dx \right| = \left| \int_{|x| \geq k} g u \rho^2 dx \right| \leq \int_{|x| \geq k} |g| |u| dx \leq \left(\int_{|x| \geq k} g^2 dx \right)^{\frac{1}{2}} \|u\|.$$

By Lemma 3.1 we find that there is a $k_2 = k_2(\epsilon)$ such that

$$\left| \int_{\mathbf{R^n}} g u \rho^2 dx \right| \leq \epsilon \quad \forall k \geq k_2 \text{ and } \forall t \geq T. \tag{4.67}$$

Note that (1.3) implies the inequality

$$\int_{\mathbf{R^n}} \rho^2 f(x,u) u dx \leq -\alpha_1 \int_{\mathbf{R^n}} \rho^2 |u|^p dx + \int_{\mathbf{R^n}} \rho^2 \phi_1(x) dx \leq \int_{|x| \geq k} |\phi_1(x)| dx.$$

Consequently, given $\epsilon \geq 0$, there exists $k_3 = k_3(\epsilon)$ such that

$$\int_{\mathbf{R^n}} \rho^2 f(x,u) u dx \leq \epsilon, \quad \forall k \geq k_3. \tag{4.68}$$

Next combine (4.63) - (4.68) and let $k_0 = max(k_1, k_2, k_3)$; then we find that

$$\frac{d}{dt}\|u\rho\|^2 + \lambda\|u\rho\|^2 \leq 3\epsilon, \quad \forall k \geq k_0 \text{ and } \forall t \geq T. \tag{4.69}$$

Therefore, by the Gronwall lemma and Lemma 3.1, we obtain the inequality, for all $k \geq k_0$ and $t \geq T$,

$$\|u(t)\rho\|^2 \leq e^{-\lambda(t-T)}\|u(T)\rho\|^2 + \frac{3\epsilon}{\lambda} \leq M^2 e^{-\lambda(t-T)} + \frac{3\epsilon}{\lambda}. \tag{4.70}$$

Therefore, if we let T_0 be sufficiently large then, $\forall t \geq T_0$ and $\forall k \geq k_0$,

$$\int_{|x|\geq k} |u(t)|^2 dx \leq \|\rho u(t)\|^2 \leq \frac{4\epsilon}{\lambda}. \tag{4.71}$$

The main result of the lemma follows from the last inequality; therefore, the proof is complete.

□

Next we derive an intermediate result needed to obtain the tail estimate for u_t in $L^2(\mathbf{R^n})$.

Lemma 4.2. *Let θ be the smooth cut-off function defined in (4.60) - (4.62) and $\rho(x) = \theta\left(\frac{|x|^2}{k^2}\right)$ with $k \geq 1$. Then there exists a positive constant c such that for any $u \in H^2(\mathbf{R^n})$,*

$$\|\rho\nabla u\| \leq c\|u\|^{\frac{1}{2}}_{H^2(\mathbf{R^n})}\|\rho u\|^{\frac{1}{2}} + \frac{c}{k}\|u\|.$$

Proof. By definition, for every $x \in \mathbf{R^n}$:

$$\nabla\rho(x) = \nabla\theta\left(\frac{|x|^2}{k^2}\right) = \theta'\left(\frac{|x|^2}{k^2}\right)\frac{2x}{k^2} \tag{4.72}$$

and

$$\Delta\rho(x) = \theta''\left(\frac{|x|^2}{k^2}\right)\frac{4|x|^2}{k^4} + \theta'\left(\frac{|x|^2}{k^2}\right)\frac{2}{k^2} \tag{4.73}$$

Note that $\theta'\left(\frac{|x|^2}{k^2}\right) = \theta''\left(\frac{|x|^2}{k^2}\right) = 0$ for any $x \in \mathbf{R^n}$ with $|x| \notin [k, \sqrt{2}k]$. Therefore, by (4.72) and (4.73) we get that

$$|\nabla\rho(x)| \leq \frac{c}{k}, \qquad \forall x \in \mathbf{R^n}, \tag{4.74}$$

$$|\Delta\rho(x)| \leq \frac{c}{k^2}, \qquad \forall x \in \mathbf{R^n}. \tag{4.75}$$

On the other hand, we have

$$\nabla(\rho u) = (\nabla\rho)u + \rho\nabla u, \tag{4.76}$$

and

$$\Delta(\rho u) = (\Delta\rho)u + 2\nabla\rho\nabla u + \rho\Delta u. \tag{4.77}$$

By (4.74)- (4.77) we find that

$$\|\nabla(\rho u)\| \leq \frac{c}{k}\|u\| + \|\nabla u\| \leq c\|u\|_{H^1} \tag{4.78}$$

and

$$\|\Delta(\rho u)\| \leq \frac{c}{k^2}\|u\| + \frac{c}{k}\|\nabla u\| + \|\Delta u\| \leq c\|u\|_{H^2(\mathbf{R^n})}. \tag{4.79}$$

By the Gagliardo-Nirenberg inequality and (4.78)-(4.79) we get that

$$\|\nabla(\rho u)\| \leq c\|\rho u\|^{\frac{1}{2}}_{H^2(\mathbf{R^n})}\|\rho u\|^{\frac{1}{2}} \leq c\|u\|^{\frac{1}{2}}_{H^2(\mathbf{R^n})}\|\rho u\|^{\frac{1}{2}},$$

which implies that

$$\|\rho\nabla u\| = \|\nabla(\rho u) - (\nabla\rho)u\|$$

$$\leq \|\nabla(\rho u)\| + \|(\nabla\rho)u\|$$

$$\leq c\|u\|^{\frac{1}{2}}_{H^2(\mathbf{R^n})}\|\rho u\|^{\frac{1}{2}} + \|(\nabla\rho)u\|. \tag{4.80}$$

By (4.74) and (4.80) we find that

$$\|\rho\nabla u\| \leq c\|u\|^{\frac{1}{2}}_{H^2(\mathbf{R^n})}\|\rho u\|^{\frac{1}{2}} + \frac{c}{k}\|u\|,$$

which completes the proof. □

Again we derive an intermediate result needed to obtain the tail estimate for $u_t(x,t)$ in $L^2(\mathbf{R^n})$. In this case, we need to derive a tail estimate for $u(x,t)$ in $H^1(\mathbf{R^n})$.

Lemma 4.3. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then given $\epsilon > 0$, there are $k_0 = k_0(\epsilon)$ and $T_0 = T_0(\epsilon)$ such that the solution satisfies*

$$\int_{|x|\geq k_0} |\nabla u(x,t)|^2 dx \leq \epsilon, \qquad \forall t \geq T_0,$$

Where k_0 depends on the data (λ, f, g), ϵ and R when $\|u_0\| \leq R$.

Proof. By Lemma 3.1, 3.6 and 4.2, we find that there is a $T_1 > 0$ such that for all $t \geq T_1$,

$$\begin{aligned}\|\rho\nabla u(t)\| &\leq c\|u(t)\|^{\frac{1}{2}}_{H^2(\mathbf{R^n})}\|\rho u(t)\|^{\frac{1}{2}} + \frac{c}{k}\|u(t)\| \\ &\leq c\|\rho u(t)\|^{\frac{1}{2}} + \frac{c}{k}. \end{aligned} \tag{4.81}$$

Note that $\rho(x) = \theta\left(\frac{|x|^2}{k^2}\right)$. By (4.81) we have, for all $t \geq T_1$,

$$\begin{aligned}\int_{\mathbf{R^n}} \theta^2\left(\frac{|x|^2}{k^2}\right)|\nabla u(x,t)|^2\,dx &\leq c\left(\int_{\mathbf{R^n}} \theta^2\left(\frac{|x|^2}{k}\right)|u(x,t)|^2 dx\right)^{\frac{1}{2}} + \frac{c}{k^2} \\ &\leq c\left(\int_{|x|\geq k} |u(x,t)|^2 dx\right)^{\frac{1}{2}} + \frac{c}{k^2}. \end{aligned} \tag{4.82}$$

Given $\epsilon > 0$, by Lemma 4.1 we find that there are $k_1 = k_1(\epsilon)$ and $T_2 = T_2(\epsilon)$ such that for all $k \geq k_1$ and $t \geq T_2$,

$$c\left(\int_{|x|\geq k} |u(x,t)|^2 dx\right)^{\frac{1}{2}} + \frac{c}{k^2} \leq \epsilon. \tag{4.83}$$

Let $T_3 = max(T_1, T_2)$. Then for all $k \geq k_1$ and $t \geq T_3$, we find from (4.82) - (4.83) that

$$\int_{|x|\geq\sqrt{2}k} |\nabla u(x,t)|^2 dx \leq \int_{\mathbf{R^n}} \theta^2\left(\frac{|x|^2}{k^2}\right) |\nabla u(x,t)|^2 dx \leq \epsilon,$$

which completes the proof. □

Finally, we derive tail estimates for u_t in $L^2(\mathbf{R^n})$.

Lemma 4.4. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then given $\epsilon \geq 0$, there are $k_0 = k_0(\epsilon)$ and $T_0 = T_0(\epsilon)$ such that the solution u satisfies*

$$\int_{|x|\geq k_0} |u_t(x,t)|^2 dx \leq \epsilon, \quad \forall t \geq T_0,$$

where k_0 depends on the data (λ, f, g) and ϵ; T_0 depends on (λ, f, g), ϵ and R when $\|u_0\| \leq R$.

Proof. Let $\rho(x) = \theta(\frac{|x|^2}{k^2})$ and take the inner product of (3.12) with $\rho^2 u_t$ in $L^2(\mathbf{R^n})$ to find that

$$\|\rho u_t\|^2 = \int_{\mathbf{R^n}} (\Delta u)\rho^2 u_t dx - \lambda \int_{\mathbf{R^n}} \rho^2 u u_t dx + \int_{\mathbf{R^n}} f(x,u)\rho^2 u_t dx + \int_{\mathbf{R^n}} g(x)\rho^2 u_t dx. \tag{4.84}$$

We now estimate every term on the right hand side of the previous equation. For the first term, we have

$$\int_{\mathbf{R^n}} (\Delta u)\rho^2 u_t dx = -\int_{\mathbf{R^n}} \nabla u((\nabla\rho^2)u_t + \rho^2 \nabla u_t) dx$$

$$
\begin{aligned}
&\leq 4\int_{\mathbf{R^n}} |\nabla u||\theta\left(\frac{|x|^2}{k^2}\right)\theta'\left(\frac{|x|^2}{k^2}\right)\frac{x}{k^2}||u_t|dx + \int_{\mathbf{R^n}} \rho|\nabla u||\nabla u_t|dx \\
&\leq 4\int_{k\leq|x|\leq\sqrt{2}k} |\nabla u||\theta\left(\frac{|x|^2}{k^2}\right)\theta'\left(\frac{|x|^2}{k^2}\right)\frac{x}{k^2}||u_t|dx + \|\rho\nabla u\|\|\nabla u_t\| \\
&\leq \frac{c}{k}\int_{k\leq|x|\leq\sqrt{2}k} |\nabla u||u_t|dx + \|\rho\nabla u\|\|\nabla u_t\| \\
&\leq \frac{c}{k}\|\nabla u\|\|u_t\| + \|\rho\nabla u\|\|\nabla u_t\|. \qquad (4.85)
\end{aligned}
$$

By Lemmas 3.2, 3.3, and 3.7, there is a $T_1 \geq 0$ such that for all $t \geq T_1$, the right-hand side of (4.85) is bounded by $\frac{c}{k} + c\|\rho\nabla u\|$ and hence we find that, for $t \geq T_1$,

$$
\begin{aligned}
\int_{\mathbf{R^n}} (\Delta u)\rho^2 u_t dx &\leq \frac{c}{k} + c\left(\int_{\mathbf{R^n}} \theta|\left(\frac{|x|^2}{k^2}\right)\nabla u|^2 dx\right)^{\frac{1}{2}} \\
&\leq \frac{c}{k} + c\left(\int_{|x|\geq k} |\nabla u|^2 dx\right)^{\frac{1}{2}}. \qquad (4.86)
\end{aligned}
$$

By (4.86) and Lemma 4.3, there is a $k_1 = k_1(\epsilon)$ and $T_2 = T_2(\epsilon)$ such that for all $k \geq k_1$ and $t \geq T_2$,

$$
\int_{\mathbf{R^n}} (\Delta u)\rho^2 u_t dx \leq \epsilon. \qquad (4.87)
$$

For the second term on the right hand side of (4.84), we have

$$
\lambda\left|\int_{\mathbf{R^n}} \rho^2 u u_t dx\right| \leq \lambda\|\rho u\|\|\rho u_t\| \leq \frac{1}{4}\|\rho u_t\|^2 + \lambda^2\|\rho u\|^2 \leq \frac{1}{4}\|\rho u_t\|^2 + \int_{|x|\geq k} |u(x,t)|^2 dx. \qquad (4.88)
$$

By (4.88) and Lemma 4.1, there is a $k_2 = k_2(\epsilon)$ and $T_3 = T_3(\epsilon)$ such that

$$
\lambda\left|\int_{\mathbf{R^n}} \rho^2 u u_t dx\right| \leq \frac{1}{4}\|\rho u_t\|^2 + \epsilon. \qquad (4.89)
$$

For the nonlinear term, using (1.4) and Young's inequality we find that

$$
\left|\int_{\mathbf{R^n}} f(x,u)\rho^2 u_t dx\right| \leq \int_{\mathbf{R^n}} |f(x,u)|\rho^2|u_t|dx
$$

$$\leq \alpha_2 \int_{\mathbf{R^n}} |u|^{p-1}\rho^2|u_t|dx + \int_{\mathbf{R^n}} \phi_2\rho^2|u_t|dx$$

$$\leq \alpha_2 \int_{\mathbf{R^n}} \rho|u|\ |u|^{p-2}\ |u_t|dx + \int_{\mathbf{R^n}} \phi_2\rho^2|u_t|dx$$

$$\leq \alpha_2\|\rho u\| \left(\int_{\mathbf{R^n}} |u|^{2p-4}\ |u_t|^2dx\right)^{\frac{1}{2}} + \int_{\mathbf{R^n}} \phi_2\rho^2|u_t|dx$$

$$\leq \alpha_2\|\rho u\|\|u\|_{2p-2}^{p-2}\|u_t\|_{2p-2} + \int_{\mathbf{R^n}} \phi_2\rho^2|u_t|dx.$$

By Lemmas 3.4 and 3.5, we get from the above that

$$\left|\int_{\mathbf{R^n}} f(x,u)\rho^2u_tdx\right| \leq c\|\rho u\| + \int_{\mathbf{R^n}} \phi_2\rho^2|u_t|dx$$

$$\leq c\left(\int_{\mathbf{R^n}} \rho^2|u(x,t)|^2dx\right)^{\frac{1}{2}} + \frac{1}{4}\|\rho u_t\|^2 + \int_{\mathbf{R^n}} \rho^2(x)\phi_2^2(x)dx$$

$$\leq \frac{1}{4}\|\rho u_t\|^2 + \left(\int_{|x|\geq k} |u(x,t)|^2dx\right)^{\frac{1}{2}} + \int_{|x|\geq k} \phi_2^2(x)dx. \tag{4.90}$$

Since $\phi_2 \in L^2(\mathbf{R^n})$, by Lemma 4.1, there is a $k_3 = k_3(\epsilon)$ and $T_4 = T_4(\epsilon)$ such that for all $k \geq K_3$ and $t \geq T_4$,

$$\left|\int_{\mathbf{R^n}} f(x,u)\rho^2u_tdx\right| \leq \int_{\mathbf{R^n}} |f(x,u)|\rho^2|u_t|dx \leq \frac{1}{4}\|\rho u_t\|^2 + \epsilon. \tag{4.91}$$

For the last term on the right hand side of (4.84), Young's inequality and Hölder's inequality imply that

$$\left|\int_{\mathbf{R^n}} g(x)\rho^2(x)u_t(x,t)dx\right| \leq \|\rho g\|\,\|\rho u_t\| \leq \frac{1}{4}\|\rho u_t\|^2 + \int_{|x|\geq k} g^2(x)dx.$$

Since $g \in L^2(\mathbf{R^n})$, there is a $k_4 = k_4(\epsilon)$ such that for all $k \geq k_4$,

$$\left|\int_{\mathbf{R^n}} g(x)\rho^2(x)u_t(x,t)dx\right| \leq \frac{1}{4}\|\rho u_t\|^2 + \epsilon. \tag{4.92}$$

Now we let $k_5 = \max\{k_1, k_2, k_3, k_4\}$ and $T_5 = \max\{T_1, T_2, T_3, T_4\}$. Then by (4.84), (4.87), (4.89), (4.91), and (4.92) we find that, for all $k \geq k_5$ and $t \geq T_5$,

$$\frac{1}{4}\|\rho u_t\|^2 \leq 4\epsilon,$$

which implies that, for $k \geq k_5$ and $t \geq T_5$,

$$\int_{|x|\geq\sqrt{2}k} |u_t(x,t)|^2 dx \leq \|\rho u_t\|^2 \leq 16\epsilon.$$

The main result of the lemma follows from the last inequality; therefore, the proof is completed. □

We have now finished deriving all of the necessary tail estimates needed to prove the asymptotic compactness of the solution operator in $H^2(\mathbf{R^n})$. In the next section, we define a semigroup for problem (1.1)-(1.2) and then use the estimates derived in this section and the previous section to prove the asymptotic compactness of resulting semigroup.

5 ASYMPTOTIC COMPACTNESS

In this section, we study the asymptotic compactness of solutions for problem (1.1)-(1.2). We first associate a semigroup with the solution operators of (1.1)-(1.2). As discussed in [5], we can prove that problem (1.1)-(1.2) is well-posed in $L^2(\mathbf{R^n})$ under assumptions (1.3)-(1.7), that is, for every $u_0 \in L^2(\mathbf{R^n})$, problem (1.1)-(1.2) has a unique solution u such that $u \in C([0,\infty), L^2(\mathbf{R^n}))$. Furthermore, the solution is continuous with respect to u_0 in $L^2(\mathbf{R^n})$. Then we can define a semigroup $\{S(t)\}_{t\geq 0}$ on $L^2(\mathbf{R^n})$ such that for every $t \geq 0$, $S(t)u_0 = u(t)$, the state of the solution of problem (1.1)-(1.2) with initial condition u_0. Notice that Lemmas 3.5 and 3.6 also show that this semigroup is actually an $(L^2(\mathbf{R^n}), L^{2p-2}(\mathbf{R^n}))$-semigroup and an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-semigroup, respectively. We now prove that this semigroup is $(L^2(\mathbf{R^n}), L^2(\mathbf{R^n}))$-asymptotically compact.

Lemma 5.1. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the semigroup $\{S(t)\}_{t\geq 0}$ is $(L^2(\mathbf{R^n}), L^2(\mathbf{R^n}))$-asymptotically compact, that is, if $\{u_{0,n}\}_{n=1}^{\infty}$ is bounded in $L^2(\mathbf{R^n})$ and $t_n \to \infty$, then $\{S(t_n)u_{0,n}\}_{n=1}^{\infty}$ has a convergent subsequence in $L^2(\mathbf{R^n})$.*

Proof. Since $\{u_{0,n}\}_{n=1}^{\infty}$ is bounded in $L^2(\mathbf{R^n})$, there is an $R > 0$ such that

$$\|u_{0,n}\| \leq R, \qquad \forall n = 1, 2, ...$$

By Lemma 3.1 we find that there is a $T_1 = T_1(R)$ such that for all $t \geq T_1$

$$\|S(t)u_{0,n}\| \leq c, \qquad \forall n = 1, 2, ...$$

Moreover, since $t_n \to \infty$, there exists $N_1 = N_1(R)$ such that $t_n \geq T_1$ for all $n \geq N_1$. So we find that for $n \geq N_1$,

$$\|S(t_n)u_{0,n}\| \leq c. \tag{5.93}$$

By Lemma 4.1, given $\epsilon > 0$, there is $k_1 = k_1(\epsilon)$ and $T_2 = T_2(\epsilon, R)$ such that for all $t \geq T_2$,

$$\int_{|x|\geq k} |S(t)u_{0,n}|^2 dx \leq \epsilon, \qquad \forall n = 1, 2, ...$$

Additionally, since $t_n \to \infty$, there exists a $N_2 = N_2(\epsilon, R)$ such that $t_n \geq T_2$ for $n \geq N_2$ and hence we obtain

$$\int_{|x|\geq k} |S(t_n)u_{0,n}|^2 dx \leq \epsilon, \qquad \forall n \geq N_2. \tag{5.94}$$

Similarly, by Lemma 3.2 and (5.93) we find that there exists $N_3 = N_3(R)$ such that for all $n \geq N_3$,

$$\|S(t_n)u_{0,n}\|_{H^1(\Omega_{k_1})} \leq c,$$

where $\Omega_{k_1} = \{x \in \mathbf{R^n} | |x| \leq k_1\}$. So by the compactness of the embedding $H^1(\Omega_{k_1}) \hookrightarrow L^2(\Omega_{k_1})$ we see that, up to a subsequence, $\{S(t_n)u_{0,n}\}_{n=1}^{\infty}$ is convergent in $L^2(\Omega_{k_1})$. Next note that

$$\begin{aligned}
\|S(t_n)u_{0,n} - S(t_m)u_{0,m}\|^2_{L^2(\mathbf{R^n})} &= \|S(t_n)u_{0,n} - S(t_m)u_{0,m}\|^2_{L^2(\Omega_{k_1})} \\
&+ \|S(t_n)u_{0,n} - S(t_m)u_{0,m}\|^2_{L^2(\mathbf{R^n}\setminus\Omega_{k_1})} \\
&\leq \|S(t_n)u_{0,n} - S(t_m)u_{0,m}\|^2_{L^2(\Omega_{k_1})} \\
&+ 2\|S(t_n)u_{0,n}\|^2_{L^2(\mathbf{R^n}\setminus\Omega_{k_1})} + 2\|S(t_m)u_{0,m}\|^2_{L^2(\mathbf{R^n}\setminus\Omega_{k_1})}.
\end{aligned} \tag{5.95}$$

Finally, we find that (5.94), (5.95) and the convergence of $\{S(t_n)u_{o,n}\}_{n=1}^{\infty}$ in $L^2(\Omega_{k_1})$ together imply that for any $\epsilon > 0$ there exists $N_4 = N_4(R, \epsilon)$ such that for all $n \geq N_4$,

$$\|S(t_n)u_{0,n} - S(t_m)u_{0,m}\|^2_{L^2(\mathbf{R^n})} \leq 5\epsilon,$$

which completes the proof. □

By the same argument as Lemma 5.1, using Lemmas 3.7 and 4.4 we can prove the following asymptotic compactness for u_t. The details are omitted here.

Lemma 5.2. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. If $\{u_{0,n}\}_{n=1}^{\infty}$ is bounded in $L^2(\mathbf{R^n})$ and $t_n \to \infty$, then $\{u_t(t_n, u_{0,n})\}_{n=1}^{\infty}$ has a convergent subsequence in $L^2(\mathbf{R^n})$, where $u(t, u_{0,n}) = S(t)u_{0,n}$.*

We next prove the asymptotic compactness of the semigroup in $L^{2p-2}(\mathbf{R^n})$. The result is needed to prove the convergence of the nonlinear term in the proof of the asymptotic compactness of the semigroup in $H^2(\mathbf{R^n})$. We first recall the following result, Lemma 2.10, in [25].

Lemma 5.3. *Let Ω be an open subset of $\mathbf{R^n}$ (bounded or unbounded). Suppose $\phi_n, \phi \in L^s(\Omega) \bigcap L^r(\Omega)$ with $1 \leq r \leq s < \infty$, $\phi_n \to \phi$ in $L^r(\Omega)$, and ϕ_n is bounded in $L^s(\Omega)$. Then $\phi_n \to \phi$ in $L^s(\Omega)$ if and only if*

$$\limsup_{m\to\infty} \limsup_{n\to\infty} \int_{\Omega_{m,n}} |\phi_n(x)|^s dx = 0,$$

where $\Omega_{m,n} = \{x \in \Omega : |\phi_n(x)| \geq m\}$.

Lemma 5.4. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. If $\{u_{0,n}\}_{n=1}^{\infty}$ is bounded in $L^2(\Omega)$ and $t_n \to \infty$, then*

$$\limsup_{m\to\infty} \limsup_{n\to\infty} \int_{\Omega_{m,n}} |S(t_n)u_{0,n}|^{2p-2} dx = 0,$$

where $\Omega_{m,n} = \{x \in \mathbf{R^n} : |S(t_n)u_{0,n}| \geq m\}$.

Proof. The proof is essentially the same as Lemma 5.4 in [25], and hence omitted here. □

From Lemmas 5.1, 5.3 and 5.4, the asymptotic compactness of solutions in $L^{2p-2}(\mathbf{R^n})$ follows immediately.

Lemma 5.5. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the semigroup $\{S(t)\}_{t\geq 0}$ is $(L^2(\mathbf{R^n}), L^{2p-2}(\mathbf{R^n}))$ - asymptotically compact, that is, if $\{u_{0,n}\}_{n=1}^{\infty}$ is bounded in $L^2(\mathbf{R^n})$ and $t_n \to \infty$, then $\{S(t_n)u_{0,n}\}_{n=1}^{\infty}$ has a convergent subsequence in $L^{2p-2}(\mathbf{R^n})$.*

We now prove the asymptotic compactness of the semigroup, $\{S(t)\}_{t\geq 0}$, in $H^2(\mathbf{R^n})$. The next proof is an adaptation of a similar proof in [25] for bounded domains.

Lemma 5.6. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the semigroup $\{S(t)\}_{t\geq 0}$ is $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$ - asymptotically compact, that is, if $\{u_{0,n}\}_{n=1}^{\infty}$ is bounded in $L^2(\mathbf{R^n})$ and $t_n \to \infty$, then $\{S(t_n)u_{0,n}\}_{n=1}^{\infty}$ has a convergent subsequence in $H^2(\mathbf{R^n})$.*

Proof. First we solve (1.1) for Δu to obtain,

$$\Delta u = \lambda u + u_t - f(x, u) - g.$$

Let $u_n(t) = S(t)u_{o,n}$. Then we have

$$\Delta(u_n(t_n)) = \lambda u_n(t_n) + u_{n,t}(t_n) - f(x, u_n(t_n)) - g. \tag{5.96}$$

By Lemmas 5.1, 5.2 and 5.5, we find that there are $u, v \in L^2(\mathbf{R^n})$ such that, up to a subsequence,

$$u_n(t_n) \to u \quad \text{in} \quad L^2(\mathbf{R^n}) \quad \text{and} \quad L^{2p-2}(\mathbf{R^n}), \tag{5.97}$$

and

$$u_{n,t}(t_n) \to v \quad \text{in} \quad L^2(\mathbf{R^n}). \tag{5.98}$$

For the nonlinear term we have

$$\begin{aligned}
\int_{\mathbf{R^n}} |f(x, u_n(t_n)) - f(x, u)|^2 dx &= \int_{\mathbf{R^n}} |\frac{\partial}{\partial u} f(x, \xi)|^2 |u_n(t_n) - u|^2 dx \\
\leq c \int_{\mathbf{R^n}} |u_n(t_n) - u|^2 dx &+ c \int_{\mathbf{R^n}} (|u_n(t_n)|^{2p-4} + |u|^{2p-4}) |u_n(t_n) - u|^2 dx \\
&\leq c\|u_n(t_n) - u\|^2 + c(\|u_n(t_n)\|_{2p-2}^{2p-4} + \|u\|_{2p-2}^{2p-4}) \|u_n(t_n) - u\|_{2p-2}^2 \\
&\leq c\|u_n(t_n) - u\|^2 + c\|u_n(t_n) - u\|_{2p-2}^2.
\end{aligned}$$

Hence, by (5.97) we obtain,

$$f(x, u_n(t_n)) \to f(x, u) \text{ in } L^2(\mathbf{R^n}). \tag{5.99}$$

By (5.96) - (5.99) we find that

$$\Delta u_n(t_n) \to \lambda u + v - f(x, u) - g \text{ in } L^2(\mathbf{R^n}), \tag{5.100}$$

which completes the proof. □

We have now finished deriving all of the necessary results related to asymptotic compactness. In the next section, the lemmas in this section will be used in conjunction with Theorem 2.7 to prove the existence of global attractors for $\{S(t)\}_{t \geq 0}$.

6 MAIN RESULTS

We are now in a position to prove the main results outlined in the abstract of this paper. In this section, we use Theorem 2.7 along with Lemmas 3.1 3.6, 5.1 and 5.6 to deduce the existence of an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor for $\{S(t)\}_{t\geq 0}$. Before we derive the main result, we prove the existence of two related attractors. First, we prove the existence of an $(L^2(\mathbf{R^n}), L^2(\mathbf{R^n}))$-global attractor for $\{S(t)\}_{t\geq 0}$.

Theorem 6.1. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the semigroup associated with the solution of (1.3) - (1.7), $\{S(t)\}_{t\geq 0}$ has an $(L^2(\mathbf{R^n}), L^2(\mathbf{R^n}))$-global attractor, which is compact in $L^2(\mathbf{R^n})$, invariant, and attracts every bounded subset of $L^2(\mathbf{R^n})$ with respect to the norm of $L^2(\mathbf{R^n})$.*

Proof. First note that Lemma 3.1 shows that there exists an $(L^2(\mathbf{R^n}), L^2(\mathbf{R^n}))$-bounded absorbing set for $\{S(t)\}_{t\geq 0}$. Additionally, by Lemma 5.1 we find that $\{S(t)\}_{t\geq 0}$ is $(L^2(\mathbf{R^n}),\ L^2(\mathbf{R^n}))$-asymptotically compact. So we find that the conditions of Remark 2.1 have been satisfied; therefore, the proof is completed. □

The next result is concerned with the existence of a global attractor in $L^{2p-2}(\mathbf{R^n})$.

Theorem 6.2. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then the semigroup associated with the solution of (1.3) - (1.7), $\{S(t)\}_{t\geq 0}$ has an $(L^2(\mathbf{R^n}), L^{2p-2}(\mathbf{R^n}))$-global attractor, which is compact in $L^{2p-2}(\mathbf{R^n})$, invariant, and attracts every bounded subset of $L^2(\mathbf{R^n})$ with respect to the norm of $L^{2p-2}(\mathbf{R^n})$.*

Proof. First note that Lemma 3.5 shows that there exists an $(L^2(\mathbf{R^n}), L^{2p-2}(\mathbf{R^n}))$-bounded absorbing set for $\{S(t)\}_{t\geq 0}$. Additionally, by Lemma 5.5 we find that $\{S(t)\}_{t\geq 0}$ is $(L^2(\mathbf{R^n}),\ L^{2p-2}(\mathbf{R^n}))$-asymptotically compact. So we find that the conditions for Theorem 2.7 have been satisfied; therefore, the proof is complete.

□

We are now in a position to prove our main result: the existence of an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor for problem (1.1)-(1.2).

Theorem 6.3. *Suppose $g \in L^2(\mathbf{R^n})$ and (1.3) - (1.7) hold. Then problem (1.1) - (1.2) has an $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-global attractor, which is compact in $H^2(\mathbf{R^n})$, invariant, and attracts every bounded subset of $L^2(\mathbf{R^n})$ with respect to the norm of $H^2(\mathbf{R^n})$.*

Proof. First note that Lemmas 3.6 and 3.1 show that there exists a bounded $(L^2(\mathbf{R^n}), H^2(\mathbf{R^n}))$-absorbing set for $\{S(t)\}_{t\geq 0}$. Additionally, by Lemma 5.6 we find that $\{S(t)\}_{t\geq 0}$ is $(L^2(\mathbf{R^n}),\ H^2(\mathbf{R^n}))$-asymptotically compact. So we find that the conditions for Theorem 2.7 have been satisfied; therefore, the proof is completed. □

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